MILITARY SPECIFICATION

LENS, DIPLOPIA TEST, RED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a red plastic lens suitable for use in the detection of diplopia.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-566	Box, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Shipping, Fiberboard.
PPP-B-676	Boxes, Setup.

Military

MIL-L-10547	Liner, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

FSC 6515

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The item shall be fabricated from Rohm and Haas transparent, red Plexiglas No. 2423, or equal. The color shall be uniform throughout.

3.1.1 Finish. The lens shall have a smooth finish, free from warpage, scratches, nicks, burrs and flash.

3.2 Design and construction. The configuration of the lens shall be in accordance with Figure 1 and shall be of single piece construction. All curves shall be smooth flowing and all edges chamfered.

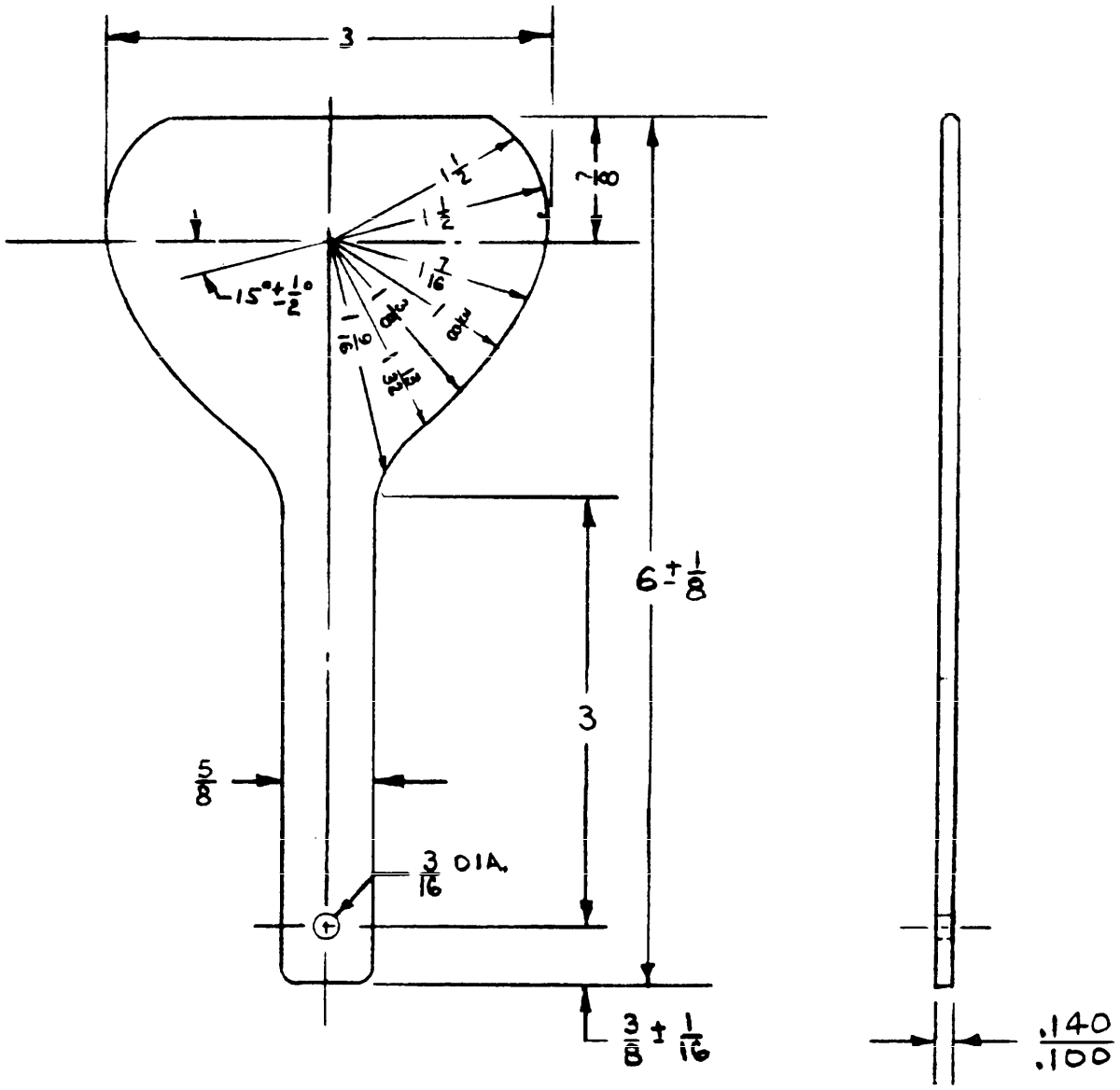
3.3 Performance.

3.3.1 Light transmission. When tested as specified in 4.4.1, the lens shall have transmission factors within the range specified in Table I at the wavelengths shown.

Table I. Lens transmission factors

Wavelength (nanometers)	Transmission range (percent)
380	6.2 - 13.4
400	1.5 - 4.8
420	0
440	0
460	0
480	0
500	0
520	0
540	0
560	0
580	0
600	2.4 - 6.9
620	25.0 - 36.3
640	62.2 - 69.6
660	83.2 - 85.7
680	89.9 - 90.5
700	91.6 - 91.8

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NOTES:
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES, TOLERANCES: ± 1/32

FIGURE 1. LENS CONFIGURATION

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3.4 Identification marking. The lens shall be clearly and legibly marked with the name or registered trademark of the manufacturer or contractor.

3.5 Workmanship. Workmanship shall be first class throughout. The lens shall be free from optical defects visible to the naked eye. The lens shall be free from all other defects which detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the plastic, including transmission values, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number (NSN), the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, the individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as indicated in Table II. The unit of product for examination purposes shall be one lens.

Table II. Sampling for examination

Type defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table III. The unit of product for test purposes shall be one lens.

Table III. Sampling for tests

Characteristic	Requirement paragraph	Test paragraph	Inspection level	AQL (percent defective)
Dimensions	3.2	--	S-2	2.5
Transmission factors	3.3.1	4.4.1	S-2	4.0

4.3 Examination. The lens shall be examined for the defects listed in Table IV. Examination, however, shall not be restricted to the defects listed in Table IV. Any lens in the sample containing one or more defects shall be rejected and if the number of defective lenses in any sample exceeds the acceptance number for that sample, the lot represented by that sample shall be rejected.

Table IV. Classification of defects

Classification	Defects
Major	
101	Lens not of one piece construction.
102	Lens not fabricated from plastic.
103	Lens not transparent.
104	Lens possesses sharp edges.
105	Finish not as specified.
Minor	
201	Lens not identified as specified in 3.4.

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4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Light transmission. The finished lens shall be tested for compliance with 3.3.1 by means of Spectrophotometric analysis.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit package. Each lens shall be wrapped in neutral tissue and placed into a carton or envelope. Closure shall be adequate to prevent accidental opening.

5.1.1.2 Intermediate package. Six (6) lens shall be packaged in a box of appropriate size constructed in accordance with PPP-B-566, PPP-B-676, or PPP-B-636, Class domestic. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Seventy two (72) lens shall be packed in a box designed for a Type 2 load and constructed in accordance with PPP-B-585, Class 3, Style 3, PPP-B-601, overseas type; PPP-B-621, Class 2; or PPP-B-636, Class weather-resistant, Grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof in accordance with PPP-B-636.

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5.2.2 Level B.

5.2.2.1 Exterior container. Seventy two (72) lens shall be packed in a box designed for a Type 2 load and constructed in accordance with PPP-B-585, Class 1, Style 3; PPP-B-601, domestic type; PPP-B-621, Class 1; or PPP-B-636, Class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The lens covered by this specification is intended for use in the detection of diplopia.

6.2 Ordering data. Procurement documents should specify:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.3 This specification is intended to cover the following item appearing in the Federal Supply Catalog:

National Stock No.	Item Identification
6515-00-346-2800	LENS, DIPLOPIA TEST, RED

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

(Project No. 6515-1904)
cs

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-L-37126 LENS, DIPLOPIA TEST, RED

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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